D4.1 Natural selection

D4 | Continuity and change
Ecosystems | SL

Understandings:

- D4.1.1—Natural selection as the mechanism driving evolutionary change
- D4.1.2—Roles of mutation and sexual reproduction in generating the variation on which natural selection acts
- D4.1.3—Overproduction of offspring and competition for resources as factors that promote natural selection
- D4.1.4—Abiotic factors as selection pressures
- D4.1.5—Differences between individuals in adaptation, survival and reproduction as the basis for natural selection
- D4.1.6—Requirement that traits are heritable for evolutionary change to occur
- D4.1.7—Sexual selection as a selection pressure in animal species
- D4.1.8—Modelling of sexual and natural selection based on experimental control of selection pressures

Key words:

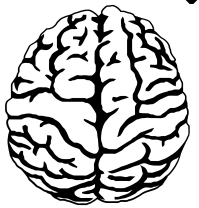
Evolution
Natural selection
Gene pool
Selection pressure
Acquired characteristics
Paradigm shift
Mutation
Genotype
Phenotype
Carrying capacity
Density dependent
Density independent
Intraspecific competition
Heritable
Sexual selection
Courtship
Guppies

IB Learner Profile

Inquirers - We nurture our curiosity, developing skills for inquiry and research. We know how to learn independently and with others. We learn with enthusiasm and sustain our love of learning throughout life.



Knowledgeable - We develop and use conceptual understanding, exploring knowledge across a range of disciplines. We engage with issues and ideas that have local and global significance.



Thinkers - We use critical and creative thinking skills to analyse and take responsible action on complex problems. We exercise initiative in making reasoned, ethical decisions



Communicators - We express ourselves confidently and creatively in more than one language and in many ways. We collaborate effectively, listening carefully to the perspectives of other individuals and groups.



Principled - We act with integrity and honesty, with a strong sense of fairness and justice, and with respect for the dignity and rights of people everywhere. We take responsibility for our actions and their consequences.



Open minded - We critically appreciate our own cultures and personal histories, as well as the values and traditions of others. We seek and evaluate a range of points of view, and we are willing to grow from the experience.



Caring - We show empathy, compassion and respect. We have a commitment to service, and we act to make a positive difference in the lives of others and in the world around us.



Risk takers - We approach uncertainty with forethought and determination; we work independently and cooperatively to explore new ideas and innovative strategies. We are resourceful and resilient in the face of challenges and change.



Balanced - We understand the importance of balancing different aspects of our lives—intellectual, physical, and emotional—to achieve well-being for ourselves and others. We recognize our interdependence with other people and with the world in which we live.

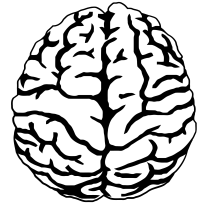


Reflective - We thoughtfully consider the world and our own ideas and experience. We work to understand our strengths and weaknesses in order to support our learning and personal development.



Guiding questions

- *What processes can cause changes in allele frequencies within a population?*
- *What is the role of reproduction in the process of natural selection?*



What is evolution?

What is evolution?

Evolution is a permanent change in the frequency of alleles in the gene pool of a population.

Evolution is caused by **natural selection**

D4.1.1—Natural selection as the mechanism driving evolutionary change



Can you give a description of evolution by natural selection?

Variation exists in population (new variants produced by mutation)

Resources are limited and often populations are too large to support

Selection pressures exists in the environment

Those with characteristics more suited survive long enough to reproduce and therefore pass on these characteristics

Those less suited die and therefore do not pass on their characteristics

The better suited characteristics become more prevalent in the population

Before Darwin many believed in “intelligent design” - that all living things were static in development and perfect in their creation.

Some tried to explain evolution in different terms:

Lamarckian evolution described evolution by **acquired characteristics**

Nature of Science

In Darwin’s time it was widely understood that species evolved, but the mechanism was not clear.

Darwin’s theory provided a convincing mechanism and replaced Lamarckism. This is an example of a **paradigm shift**. Students should understand the meaning of the term “paradigm shift”.

D4.1.2—Roles of mutation and sexual reproduction in generating the variation on which natural selection acts

Mutation

Changes to **genotype** create changes in **phenotype**

New base sequence

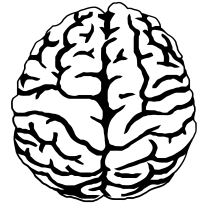
New genes

New amino acid sequences

New proteins

New characteristics

Sexual reproduction



New combinations of alleles

What are the steps in sexual reproduction that create variation?

Sexual reproduction

New combinations of alleles

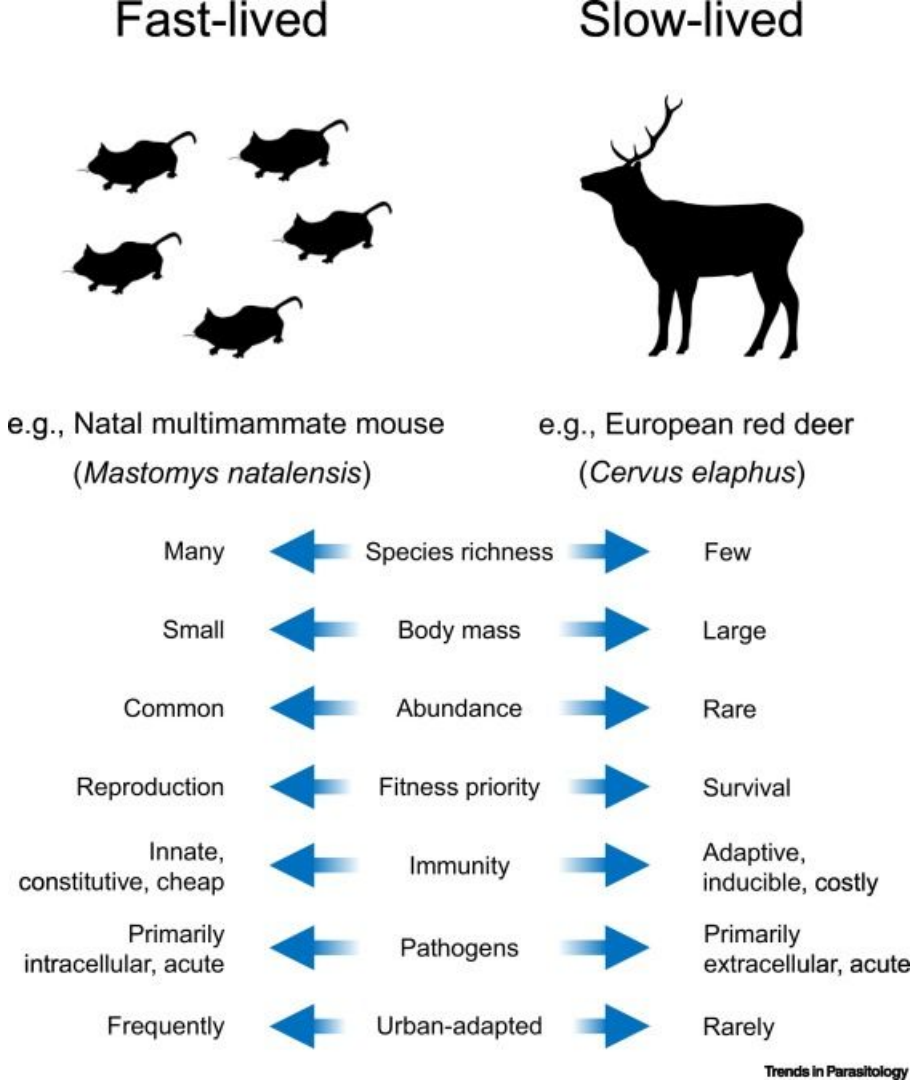
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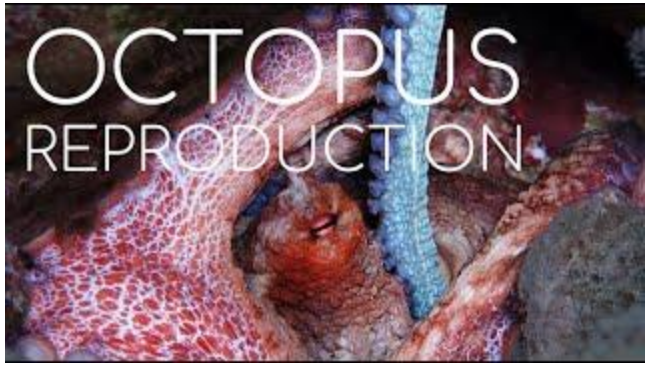
Crossing over, Independent assortment, Random fertilisation.

D4.1.3—Overproduction of offspring and competition for resources as factors that promote natural selection

Genes are more likely to be passed on if there are large numbers of offspring.

Different organisms have dramatically different reproductive rates and strategies.


Fast-lived
Slow-lived
e.g., Natal multimammate mouse
(*Mastomys natalensis*)
e.g., European red deer
(*Cervus elaphus*)
Many
Species richness
Few
Small
Body mass
Large
Common
Abundance
Rare
Reproduction
Fitness priority
Survival
Innate, constitutive, cheap
Immunity
Adaptive, inducible, costly
Primarily intracellular, acute
Pathogens
Primarily extracellular, acute
Frequently
Urban-adapted
Rarely
Trends in Parasitology


OCTOPUS
REPRODUCTION

Regardless of the method of reproduction, species tend to have more offspring than the **carrying capacity** of the environment.

Leads to competition!

Carrying capacity



List some factors that may determine the carrying capacity of an environment.

Carrying capacity

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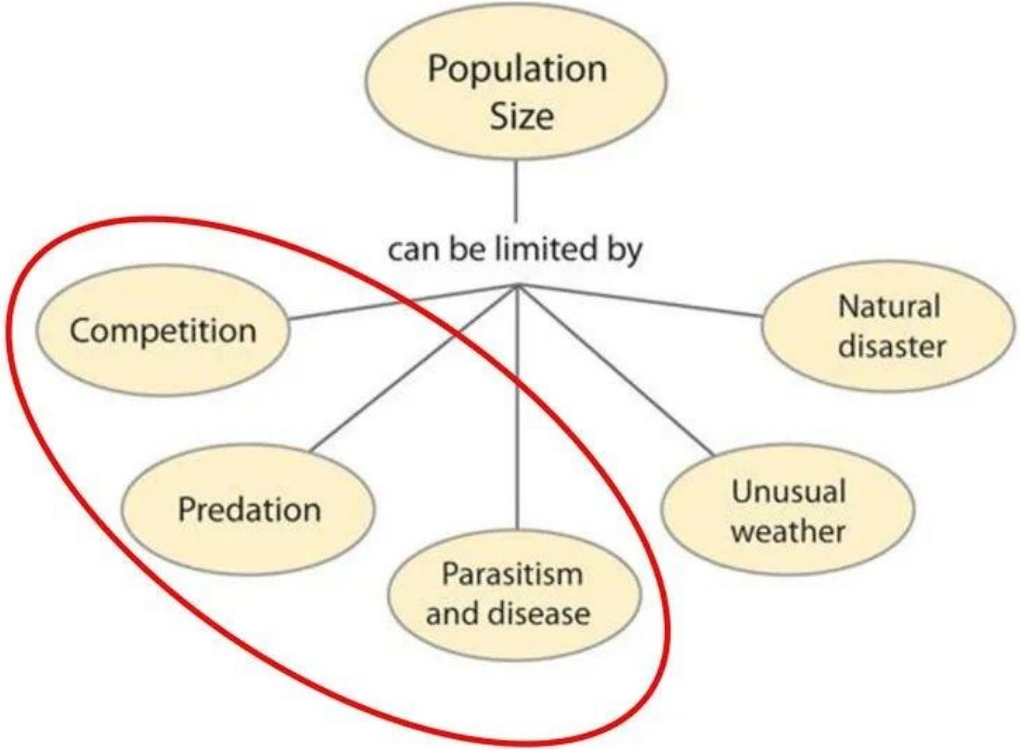
Plants: Light, water, nutrients

Animals: Food source, shelter, predators

D4.1.4—Abiotic factors as selection pressures

Density-Dependent Factors

Density-dependent limiting factors operate strongly when population density reaches a certain level.



Density dependent factors tend to be **biotic**

Density independent factors tend to be **abiotic**



Describe how an abiotic factor may act as a selection pressure

Examples

Temperature

Organisms must cope with temperature extremes. Species living in cold climates may evolve insulation mechanisms like thicker fur, while those in hot climates might develop efficient cooling mechanisms or behaviors such as burrowing.

Water Availability

In arid environments, plants might evolve deep root systems, reduced leaf surface area, or other mechanisms to conserve water. Animals might develop behaviors to reduce water loss or physiological adaptations like efficient kidneys.

In areas prone to flooding, organisms may evolve traits that allow them to survive or even thrive in waterlogged conditions.

Light Intensity

Plants in low-light environments might evolve larger leaves or different pigments to maximize light absorption. In contrast, those in high-light environments might develop mechanisms to avoid photodamage.

Animals might evolve behaviors to avoid predators or find food based on light conditions, such as nocturnal habits in desert animals.

D4.1.5—Differences between individuals in adaptation, survival and reproduction as the basis for natural selection



What does “survival of the fittest” mean to you?

What does “survival of the fittest” mean to you?

Intraspecific competition

Individuals of the same species are competing for exactly the same niche.

Those that are “fittest” are more likely to survive to be able to reproduce and therefore pass on their genes.

D4.1.6—Requirement that traits are heritable for evolutionary change to occur



Are acquired characteristics heritable?

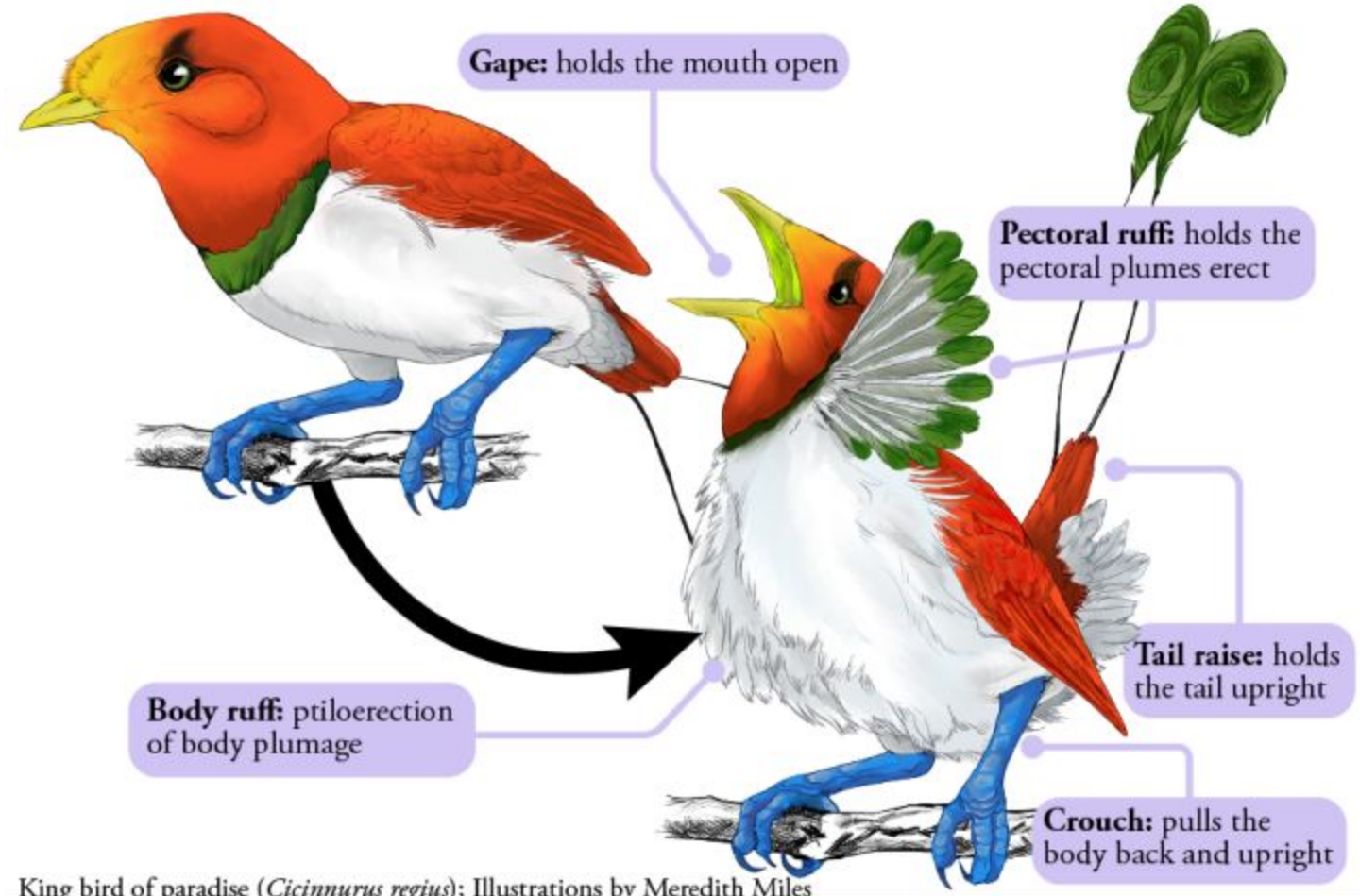
Are acquired characteristics heritable?

Characteristics that are acquired in one's lifetime (for example being able to play the piano) are not generally encoded into the genome and therefore cannot be passed on to future generations!

D4.1.7—Sexual selection as a selection pressure in animal species



Why are some animals considered “more attractive” to the opposite sex?



King bird of paradise (*Cicinnurus regius*); Illustrations by Meredith Miles



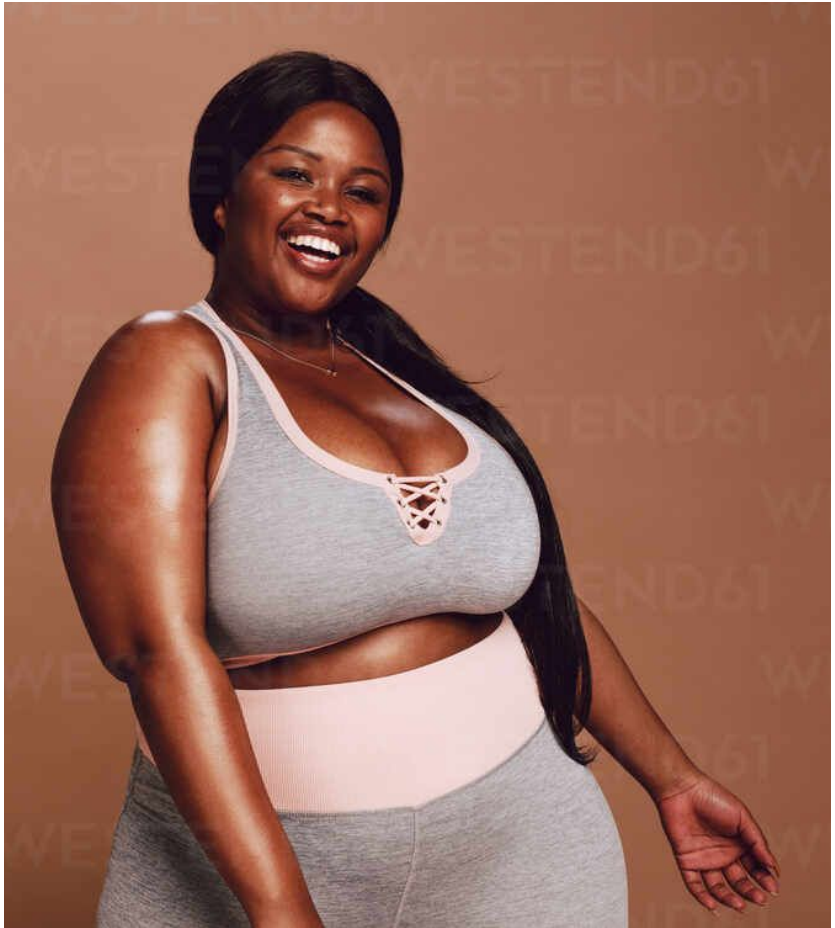
Why might an ornate and colourful male bird be considered a good “catch” for the females?

Why might an ornate and colourful male bird be considered a good “catch” for the females?

Sick or weak bird would not have the energy and nutrients that goes into growing elaborare colourful tail feathers.

Bright colours make the birds easy prey… Must be good at avoiding predators!

How might society determine what it means to be “attractive”?





Courtship

In many animal species an elaborate courtship ritual is used to determine the fitness of potential mates.





D4.1.8—Modelling of sexual and natural selection based on experimental control of selection pressures

Endler’s experiments with guppies

Endler chose small fish in Trinidad and Tobago to study evolution.

Guppies can reproduce approximately every month and can have over 100 offspring per time.



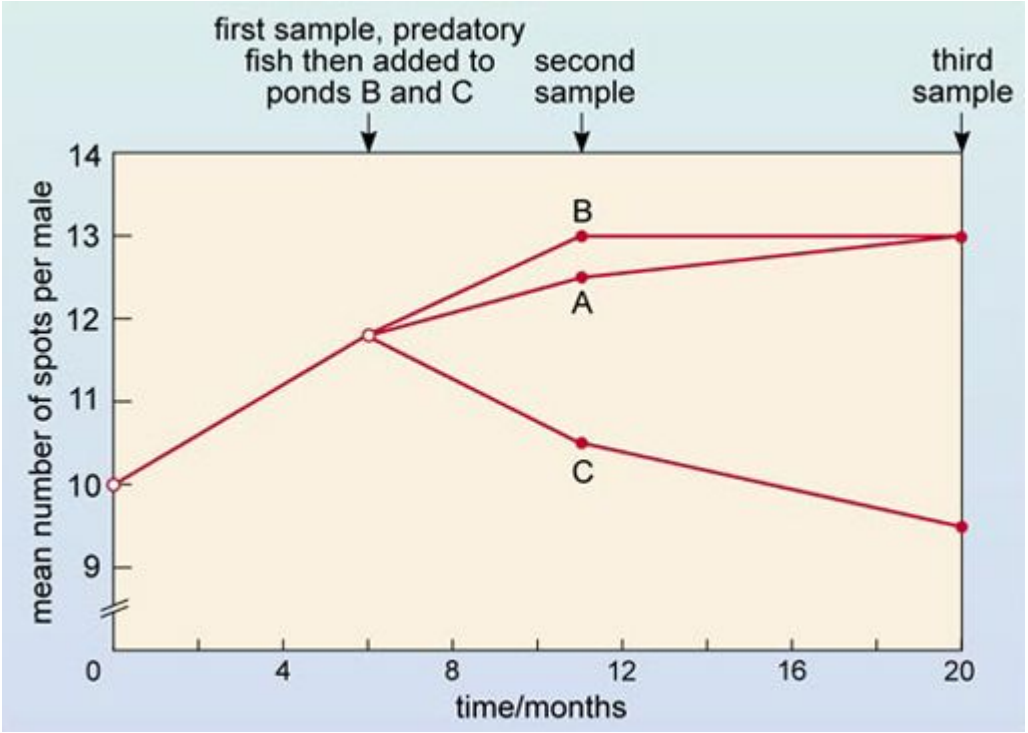
1. Kept fish in artificial tanks.
    A. No predator
    B. Added 6 *Rivulus hartii* which is not a predator of guppies
    C. Added 1 *Crenicichla alta* which is a predator of guppies

Describe, Interpret, Conclude

Describe the graph.

Interpret by putting it into context of what you have learned (about evolution by natural selection)

Conclude by summarising



A. No predator
B. Added 6 *Rivulus hartii* which is not a predator of guppies
C. Added 1 *Crenicichla alta* which is a predator of guppies

Note that a measurement of variance (eg. standard deviation) isn't given. Do you think that this would have been useful?

Does this graph say anything about the number of fish?

Conclusion

Females prefer more spots

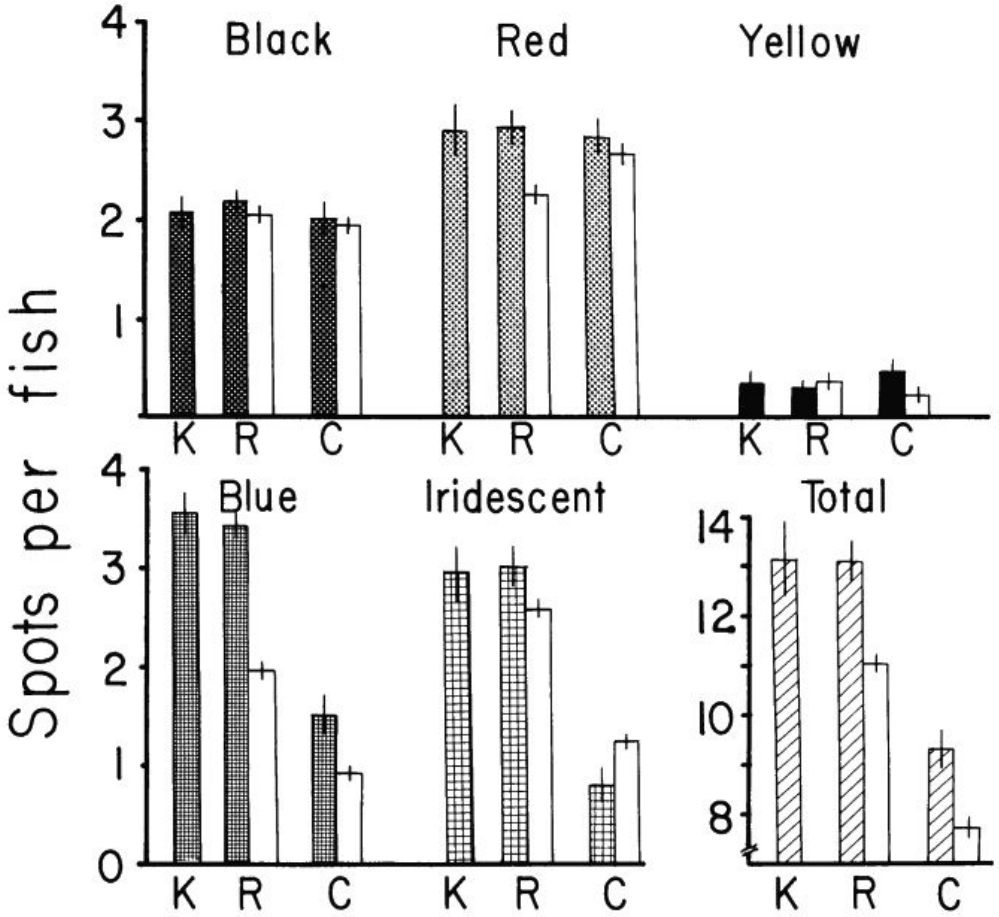
Predators prefer more spots

Experiment shows the comparative effects of **sexual selection** and **natural selection**

Shaded bars: greenhouse
Unshaded bars: field

K =ponds with no predation

R = ponds with *R. hartii (killifish)* - not a predator
C = ponds with *C. alta (cichlids)* - predator



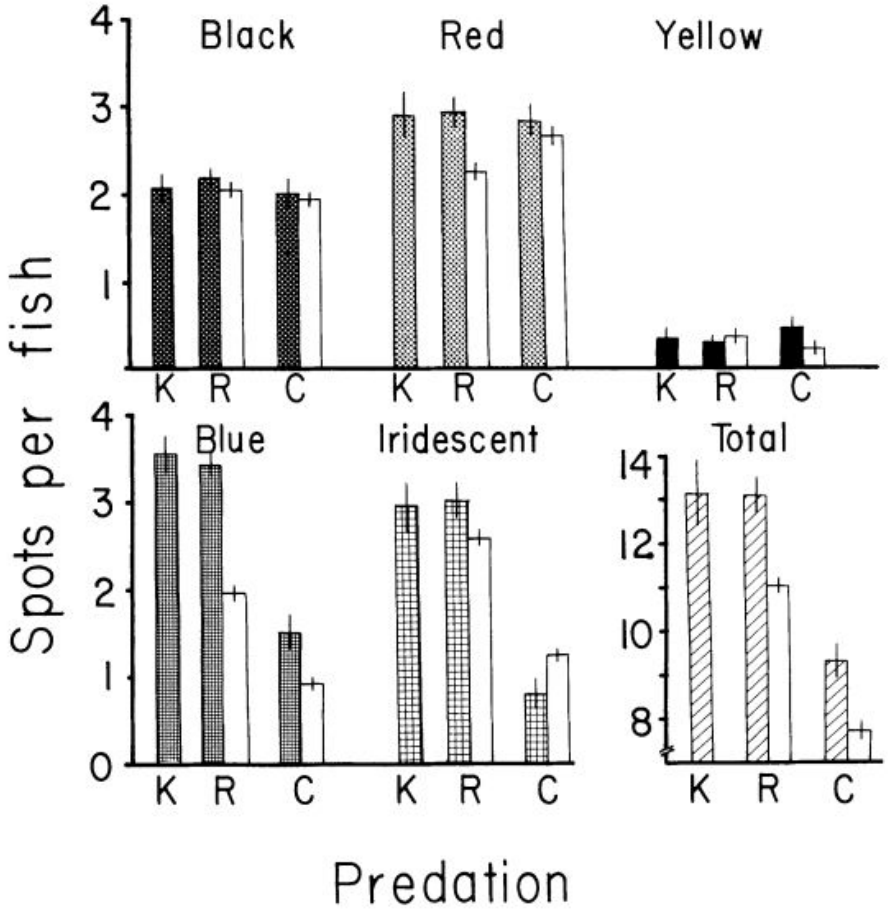
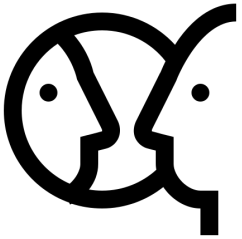


FIG. 2. Number of spots per fish in the greenhouse at Census II and in the field. Small vertical lines are two standard errors. Shaded bars: greenhouse means. Unshaded bars: field results. *K*, ponds with no predation. *R*, ponds with *R. hartii*. *C*, ponds with *C. alta*. Note similarity between field and greenhouse results, and lack of predation effect for pigment colors: black, red, and yellow.

Can you...



Define evolution in terms of the gene pool?

Describe evolution in terms of natural selection?

Explain how variation arises?

Explain how the carrying capacity of an environment can lead to evolution?

Explain the role of sexual selection in evolution?

Describe Endler’s experiment on Guppy fish?